

Validation Method Standard

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· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
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Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Canonical Definition

Validation Method Standard (VMTS) defines the governance conditions under which validation methods are identified, selected, justified, configured, combined, controlled, and maintained so that established Validation Criteria can be examined through methods appropriate to the Validation Object, Validation Purpose, Validation Scope, required Validation Depth, applicable conditions, and intended Validation Determination.

Canonical Definition Extension

- A Validation Object may be precisely established.
 - Its purpose and scope may be governed.
 - Its Validation Criteria may be explicit.
 - None of these conditions determines how those criteria should actually be examined.
 - Validation requires a method capable of producing meaningful information about whether the established criteria have been satisfied.
 - The method must fit the object.
 - It must fit the criterion.
 - It must fit the required depth.
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It must remain appropriate within the relevant operational and contextual conditions.

Validation Method Standard therefore governs the transformation of an approved **Validation Criteria Framework** into a controlled **Validation Method Framework**.

The standard establishes the conditions necessary to determine:

- which validation methods are suitable,
- why a particular method is selected,
- whether the method can actually evaluate the relevant criterion,
- which assumptions govern the method,
- which limitations constrain its use,
- whether multiple methods are required,
- how methods may be combined,
- whether method independence is necessary,
- how methodological conflicts are governed,
- how method configuration is controlled,
- and when a method must be reassessed or replaced.

VMTS does not determine whether the evidence generated or consumed by a method is sufficient.

It does not determine whether evidence sources are reliable.

It does not execute the validation itself.

It does not determine the final validity state.

VMTS governs the methodological means through which established Validation Criteria become capable of examination.

Core Question

Which validation method, or combination of methods, can legitimately and sufficiently examine the established Validation Criteria under the governed validation conditions?

Purpose

The purpose of Validation Method Standard is to ensure that validation methodology is selected because it is appropriate to the validation requirement rather than because it is convenient, familiar, inexpensive, available, or likely to produce a preferred result.

The standard prevents validation from becoming dependent upon:

- arbitrary method selection,
- habitual method reuse,
- method-object mismatch,

- method-criterion mismatch,
- unjustified methodological assumptions,
- insufficient methodological depth,
- hidden method limitations,
- inappropriate transfer of methods between domains,
- selective use of favorable methods,
- or post-hoc methodological adjustment.

VMTS establishes the methodological foundation required for controlled validation execution.

Governance Objective

VMTS establishes a governed Validation Method Framework capable of supporting methodologically appropriate, proportionate, reproducible, transparent, and traceable validation.

The standard seeks to ensure that:

- validation methods are identifiable,
 - method selection is justified,
 - methods correspond to established Validation Criteria,
 - methods are suitable for the Validation Object,
 - methods support the required Validation Depth,
 - assumptions remain explicit,
 - methodological limitations remain visible,
 - method configurations remain controlled,
 - multiple methods can be coherently combined where required,
 - method independence is preserved where necessary,
 - methodological conflicts are governable,
 - method changes remain traceable,
 - and the resulting methodological structure is fit for validation execution.
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Operational Role

Validation Method Standard serves as the **methodological design stage of the VALIDOS® validation governance lifecycle.**

The preceding standards establish:

What is being validated → Why and within what boundaries → Against which criteria

VMTS establishes:

How those criteria can legitimately be examined.

Its architectural position is:

Validation Criteria Framework → Method Requirements → Method Selection → Method Configuration → Method Integration → Method

Readiness → Validation Method Framework

The resulting Validation Method Framework becomes the governed methodological basis for subsequent evidence and execution governance.

Governance Scope

Validation Method Standard governs:

- validation-method identification,
 - method eligibility,
 - method suitability,
 - method selection,
 - method justification,
 - method-to-criterion alignment,
 - method-to-object alignment,
 - methodological assumptions,
 - methodological limitations,
 - method configuration,
 - method calibration,
 - method independence,
 - multi-method validation,
 - methodological triangulation,
 - method complementarity,
 - methodological conflict,
 - method reproducibility,
 - method repeatability,
 - method sensitivity,
 - method robustness,
 - method versioning,
 - method change,
 - method readiness,
 - and Validation Method Framework integrity.
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Validation Method Governance Model

VMTS establishes the following structural progression:

Validation Criteria Framework → Method Requirement Identification → Candidate Method Identification → Suitability Assessment → Method Selection → Method Configuration → Multi-Method Integration → Method Readiness → Validation Method Framework

This progression creates a controlled transition from evaluation conditions to the methodological mechanisms through which those conditions can be examined.

Method Requirement Identification

Before selecting a method, VMTS requires the methodological needs of the Validation Criteria Framework to be understood.

Relevant requirements may include:

- what characteristic must be examined,
- what type of observation is required,
- required precision,
- required confidence,
- required Validation Depth,
- temporal requirements,
- environmental conditions,
- population requirements,
- reproducibility requirements,
- independence requirements,
- and known Validation Limitations.

The method should follow the validation requirement.

The validation requirement should not be reshaped merely to fit the available method.

Validation Method Types

VMTS is method-neutral.

Depending upon the Validation Object and criteria, legitimate validation methods may include:

- empirical validation,
- experimental validation,
- observational validation,
- statistical validation,
- analytical validation,
- computational validation,
- documentary validation,
- comparative validation,
- benchmark validation,
- expert validation,
- consensus validation,
- formal verification,
- cryptographic validation,
- simulation-based validation,
- scenario-based validation,
- replication,
- cross-validation,
- adversarial validation,
- independent review,
- or other methodologically defensible approaches.

VALIDOS® does not assume that one validation method is universally superior.

Method legitimacy depends upon its relationship to the governed

validation requirement.

Method Suitability

A candidate method must be assessed for suitability before it becomes part of the Validation Method Framework.

Suitability may depend upon:

- Validation Object characteristics,
- Validation Criterion type,
- Validation Threshold structure,
- required Validation Depth,
- expected evidence,
- operating environment,
- population,
- uncertainty,
- complexity,
- dependencies,
- intended reliance,
- and known limitations.

A technically sophisticated method may still be unsuitable for a particular validation purpose.

Method-to-Criterion Alignment

Every material Validation Criterion should have an identifiable methodological path through which it can be examined.

VMTS therefore establishes the relationship:

Validation Criterion → Methodological Requirement → Validation Method

A method must not be assumed to validate conditions it was not designed or capable of examining.

Likewise, a Validation Criterion should not be considered methodologically covered merely because some validation activity occurred.

Method-to-Object Alignment

The method must also be appropriate to the nature of the Validation Object.

Methods appropriate for:

- a factual claim,
 - an AI-generated output,
 - a statistical model,
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- a physical component,
- a decision,
- a simulation,
- a dataset,
- an autonomous system,
- or an organizational process

may differ substantially.

VMTS prevents methodological transfer between materially different object types without sufficient justification.

Method Assumptions

Validation methods may depend upon assumptions.

These may concern:

- data distribution,
- system stability,
- independence,
- environmental conditions,
- population characteristics,
- causal relationships,
- model behavior,
- measurement reliability,
- sampling,
- or other methodological conditions.

Material assumptions must remain explicit.

Where an assumption cannot be sufficiently supported, its effect upon methodological validity must be governed.

Method Limitations

Every validation method has boundaries.

A method may be unable to:

- observe certain behaviors,
- represent particular environments,
- detect rare conditions,
- establish causality,
- generalize across populations,
- examine hidden dependencies,
- or support a required degree of confidence.

VMTS requires material method limitations to remain connected to the Validation Method Framework.

Methodological limitations may propagate into later evidence, determination, applicability, and reliance governance.

Method Independence

Some validations require independent methodological assessment.

Independence may concern:

- evaluator independence,
- methodological independence,
- source independence,
- tool independence,
- dataset independence,
- organizational independence,
- or independence between multiple validation methods.

VMTS governs when independence is methodologically necessary and how it should be preserved.

Multiple methods using the same underlying assumptions, data, or analytical mechanism should not automatically be represented as independent confirmation.

Multi-Method Validation

A single method may not be sufficient for every Validation Criterion.

VMTS therefore supports **Multi-Method Validation** where multiple methods are required to examine different dimensions of the Validation Object or to increase methodological robustness.

Multiple methods may be:

- complementary,
- sequential,
- parallel,
- independent,
- confirmatory,
- adversarial,
- or conditional.

The existence of multiple methods does not automatically create stronger validation.

Their relationships must be governed.

Methodological Triangulation

Where appropriate, validation may use methodologically distinct approaches to examine the same or related Validation Criteria.

Triangulation may strengthen validation where different methods provide genuinely independent or complementary perspectives.

However:

Method Count ≠ Validation Strength

Five methods sharing the same hidden assumption or evidence source may provide less independent validation than two genuinely distinct methods.

VMTS therefore governs the structure of methodological diversity rather than merely the number of methods used.

Methodological Conflict

Different methods may produce conflicting results.

VMTS does not assume that one method should automatically override another.

Methodological conflict may indicate:

- different assumptions,
- different sensitivities,
- different evidence,
- different populations,
- different operating conditions,
- methodological error,
- or genuine uncertainty.

Material conflict must remain visible for subsequent VALIDOS® governance.

Method Configuration

Many methods require configuration before execution.

Configuration may include:

- parameters,
 - sampling design,
 - test conditions,
 - benchmark selection,
 - scenario construction,
 - simulation settings,
 - confidence requirements,
 - evaluation windows,
 - environmental controls,
 - or other methodological settings.
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Material configuration choices must be documented and justified.

A method cannot be considered fully governed if its name is known but the configuration materially determining its behavior is not.

Method Calibration

Where applicable, a validation method may require calibration against:

- reference conditions,
- known outcomes,
- control samples,
- benchmark environments,
- baseline systems,
- validated instruments,
- or other methodological references.

Calibration must remain distinguishable from the validation result itself.

Method Reproducibility

Where methodologically appropriate, the Validation Method Framework should support sufficient reproducibility for another qualified evaluator to understand and, where possible, reproduce the methodological process.

Reproducibility requirements may vary by domain.

VMTS does not require every validation to be perfectly reproducible.

It requires the degree of reproducibility necessary for the governed Validation Purpose to be explicit where material.

Method Robustness

VMTS may assess whether a method remains suitable under reasonable variations in:

- inputs,
- samples,
- parameters,
- environments,
- evaluators,
- tools,
- or operating conditions.

Method robustness is particularly important where small

methodological changes could materially alter the resulting evaluation.

Method Readiness

Before validation execution begins, the Validation Method Framework should be sufficiently ready for governed use.

Method readiness may require:

- suitable methods have been identified,
- criteria-method relationships are established,
- configurations are defined,
- assumptions are explicit,
- limitations are documented,
- required independence is established,
- multi-method relationships are governed,
- necessary calibration is complete,
- and material methodological conflicts are addressed or explicitly preserved.

A method may therefore be technically available while remaining

Validation Method Not Ready.

Method Change Governance

Validation methodology may require modification because of:

- changed Validation Criteria,
- new scientific knowledge,
- method failure,
- new evidence,
- changed operating conditions,
- identified methodological limitations,
- improved validation technology,
- or discovery that the original method is unsuitable.

Material method change must be:

- identified,
- justified,
- documented,
- versioned,
- and assessed for its effect upon previous validation activity.

A changed method must not silently rewrite the methodological basis of historical validation results.

Method Versioning

Every material validation result should remain traceable to the method version and configuration under which it was produced.

This preserves the ability to answer:

How was this validation result produced?

and:

Would the same result still be methodologically comparable under the current method?

Method Selection Independence

Where stakeholders have incentives linked to the validation outcome, method selection may itself become vulnerable to bias.

VMTS therefore supports governance controls capable of separating method selection from preferred outcomes where necessary.

The method should be selected because it is capable of answering the validation question—not because it increases the probability of obtaining a desired conclusion.

Governance Boundary

Validation Method Standard begins when a sufficiently coherent Validation Criteria Framework exists and methodological means must be established for examining it.

It ends when a sufficiently justified, configured, traceable, and validation-ready Method Framework exists to support subsequent evidence and validation execution governance.

VMTS does not determine:

- the identity of the Validation Object,
- Validation Purpose or Scope,
- Validation Criteria,
- evidence fitness,
- source reliability,
- validation execution results,
- final Validation Determination,
- Validation State,
- reliance authorization,
- or revalidation requirements.

These responsibilities belong to other VALIDOS® Parent Standards.

Minimum Governance Requirements

A conforming implementation of VMTS should establish:

- 1. identifiable methodological requirements,
- 2. suitable candidate validation methods,
- 3. documented method-selection logic,
- 4. method-to-criterion alignment,
- 5. method-to-object alignment,
- 6. explicit material assumptions,
- 7. explicit material limitations,
- 8. controlled method configurations,
- 9. governed multi-method relationships where applicable,

10. independence requirements where necessary, 11. method versioning and change control, 12. and a Validation Method Framework sufficiently ready for subsequent validation execution.

Governance Outputs

VMTS may produce:

- Validation Method Framework,
- Method Requirement Record,
- Candidate Method Register,
- Method Suitability Assessment,
- Method Selection Record,
- Criteria-to-Method Map,
- Method Assumption Record,
- Method Limitation Record,
- Method Configuration Record,
- Method Calibration Record,
- Method Independence Record,
- Multi-Method Relationship Map,
- Methodological Conflict Record,
- Method Readiness Determination,
- Method Version Record,
- Method Change Record.

These outputs establish the methodological basis required by subsequent VALIDOS® governance.

Operational Applications

Validation Method Standard may be applied across:

- artificial intelligence,
- machine learning,
- autonomous systems,
- AI-generated outputs,
- claims and information,
- datasets,
- predictions,

- simulations,
- analytical models,
- decision systems,
- enterprise systems,
- financial services,
- healthcare,
- industrial operations,
- scientific environments,
- regulatory compliance,
- critical infrastructure,
- digital assets,
- digital identities,
- operational governance,
- and any governed environment requiring methodologically controlled validation.

Relationship to Other OOF Architectures

Validation Method Standard interoperates with:

- **GOA™**
- **OBIDENTITY®**
- **INTEGROS®**
- **ORA™**
- **AGA™**
- **AIG®**
- **CLIA®**
- **MGIA™**
- **ASGA™**
- **RIS™**

by providing the canonical validation-governance methodology through which requirements originating from complementary governance spaces can be examined using methods appropriate to the Validation Object, Validation Criteria, and governed validation conditions.

VMTS does not replace scientific methods, technical testing standards, audit procedures, regulatory methodologies, or domain-specific validation practices.

It governs **how such methods are selected, justified, structured, controlled, combined, and maintained within the VALIDOS® validation lifecycle.**

Foundational Principle

A validation method is legitimate only when it is fit for the criterion it is intended to examine.

Canonical Closing Statement

Validation Method Standard establishes the canonical methodological-design layer of VALIDOS® — Validation Governance Architecture. By governing method requirements, suitability, selection, justification, criterion alignment, object alignment, assumptions, limitations, configuration, calibration, independence, methodological diversity, conflict, readiness, versioning, and change, VMTS ensures that validation is performed through methods selected for their ability to examine the governed validation conditions rather than for convenience, availability, or preferred outcomes.

Validation Method Standard

Architecture: VALIDOS® — Validation Governance Architecture

Category: AI & Interpretation · **Subcategory:** Validation Governance

Canonical Definition: Validation Method Standard (VMTS) defines the governance conditions under which validation methods are identified, selected, justified, configured, combined, controlled, and maintained so that established Validation Criteria can be examined through methods appropriate to the Validation Object, Validation Purpose, Validation Scope, required Validation Depth, applicable conditions, and intended Validation Determination.

Governed Space: Validation Method

A validation method is legitimate only when it is fit for the criterion it is intended to examine.

→ [View Standard](#)

Aboit standard

Module Architecture

→ [Validation Method Eligibility Module \(VMEM\)](#)

→ [Validation Method Suitability Module \(VMSM\)](#)

→ [Validation Method Configuration Module \(VMCM\)](#)

→ [Validation Method Integration Module \(VMIM\)](#)

→ [Validation Method Readiness Module \(VMRM\)](#)

Parent Resources

→ [VALIDOS® Architecture Map](#)

→ [VALIDOS® Complete Standards & Modules Index](#)

→ [VALIDOS® — Four-Layer Validation Protocol™](#)

Standards Compatibility

→ [Validation Criteria Standard](#)

Operational compatibility within the governed validation lifecycle.

→ [Validation Evidence Fitness Standard](#)

Operational compatibility within the governed validation lifecycle.

→ [Validation Execution Standard](#)

Operational compatibility within the governed validation lifecycle.

Related Documents

→ [About Validation Method Standard](#)

→ [VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

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Canonical Source

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